

Agustín Nolasco

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🌐 Personal Website

🐙 GitHub



Education

- 2024–Present 📖 **Ph.D. Student in Computer Science** in National University of Córdoba, FAMAF. Córdoba.
- 2019–2024 📖 **Licentiate in Computer Science** in National University of Río Cuarto. Córdoba.
Final Average Score: 9.88 out of 10.
Thesis title: *Inference of Equivalent Method Sequences for Metamorphic Testing*.
- 2019–2021 📖 **Computing Analyst** in National University of Río Cuarto. Córdoba.
Final Average Score: 9.81 out of 10.

Teaching Experience

- 2024 – Jun. 2026 📖 **Teaching Assistant**, National University of Río Cuarto, Argentina
Courses: Logics and Problem Solving – Compilers – Design of Algorithms – Algorithms and Data Structures
- 2023 – 2025 📖 **Grader**, Guangdong Technion – Israel Institute of Technology, China
Courses: Seminars in Computer Science – Introduction to Computer Science – Introduction to System Programming
- 2022 – 2024 📖 **Student Teaching Assistant**, National University of Río Cuarto, Argentina
Courses: Advanced Programming – Design of Algorithms

Industrial Experience

- Apr. 2024 – Aug. 2024 📖 **QA Tester**, Oliver Informatics, Río Cuarto, Córdoba, Argentina

Research Publications

Journal Articles

- 1 A. Nolasco, F. Molina, R. Degiovanni, *et al.*, “Abstraction-aware inference of metamorphic relations,” *Proc. ACM Softw. Eng.*, vol. 1, no. FSE, Jul. 2024. 📄 DOI: 10.1145/3643747.

Conference Proceedings

- 1 A. Balestra, A. Nolasco, F. Molina, D. Garbervetsky, R. Degiovanni, and N. Aguirre, “Improving dynamic specification inference with llm-generated counterexamples,” in *2026 IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2026, pp. 393–404. 📄 DOI: 10.1109/ICST69053.2026.00059.
- 2 A. Nolasco, “Automated inference of expressive metamorphic relations for effective metamorphic testing,” in *2026 IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2026, pp. 22–24. 📄 DOI: 10.1109/ICST69053.2026.00016.

Preprints

- 1 R. Degiovanni, F. Molina, A. Nolasco, N. Aguirre, and M. Papadakis, *Specification inference for evolving systems*, 2023. arXiv: 2301.12403 [cs.SE].  URL: <https://arxiv.org/abs/2301.12403>.

Courses

- May. 2025  **Introduction to Software Testing**
taught by Dr. Andrea Bombarda
Final Score: 8 out of 10
- Apr. 2025  **Parallel Computing**
taught by Dr. Nicolás Wolovick, FAMAf, National University of Córdoba
Final Score: 10 out of 10.
- Aug. 2024  **Text Mining**
taught by Dra. Laura Alonso Alemany, FAMAf, National University of Córdoba
Final Score: 10 out of 10.
-  **Neural Networks**
taught by Dr. Francisco Tamarit, FAMAf, National University of Córdoba
Final Score: 10 out of 10
- Mar. 2025  **Advances in Automatic Software Testing**
taught by Dr. Mike Papadakis, Río 2025 Summer School
Final Score: 9 out of 10
-  **Introduction to Input Grammar Fuzzing**
taught by Dr. Andreas Zeller, Río 2025 Summer School
Final Score: 10 out of 10
-  **Analysis of mobile apps for quality and trustworthiness**
taught by Dr. Alessandra Gorla, Río 2025 Summer School
Final Score: 10 out of 10
- Feb. 2024  **Dynamic Modal Logics**
taught by Dr. Raul Fervari and Dr. Carlos Areces, Río 2024 Summer School
Final Score: 6 out of 10
-  **Verification of Fault Trees**
taught by Dr. Joost-Pieter Katoen, Río 2024 Summer School
- Aug. 2023  **Advanced Data Structures**
taught by Dr. Conrado Martínez, ECI 2023
Final Score: 10 out of 10
-  **Fuzzing for Compiler Testing**
taught by Dr. Guillermo Polito, ECI 2023
Final Score: 9 out of 10
- Feb. 2023  **Modeling and Analyzing Concurrent Systems**
taught by Dr. Thomas Noll, Río 2023 Summer School
Final Score: 8 out of 10
-  **Formal Approaches to Decision-Making under Uncertainty**
taught by Dr. Arnd Hartmanns, Río 2023 Summer School
- Aug. 2022  **Introduction to APIs**
taught by Dr. Guido De Caso, ECI 2022
Final Score: 8 out of 10
-  **Satellite Images Processing**
taught by Dr. Rafael Grimson, ECI 2022
Final Score: 7 out of 10

Courses (continued)

- Oct. 2019  **Software Testing**
taught by Dr. Nazareno Aguirre, CACIC 2019
Final Score: 10 out of 10
- 2011–2016  **English Academy**
Instituto Superior de Lengua Inglesa (ISLI), Villa Mercedes, San Luis

Research Projects

- Dec. 2021 – Present  **Optimizations to Symbolic Execution and Applications**
PICT 2019
Director: Dr. Nazareno Aguirre
Position: Student Collaborator
Funded by: National Agency for the Promotion of Research, Technological Development, and Innovation, through FONCyT.
- Jan. 2020 – Present  **Application of Constraint Solving Techniques to Software Validation and Verification**
PPI 2020-2022
Director: Dr. Nazareno Aguirre
Position: Student Collaborator
Funded by: Secretariat of Science and Technology, National University of Río Cuarto.

Research Grants and Scholarships

- Ago. 2024 – Present  **Ph.D. Scholarship.** National Scientific and Technical Research Council (CONICET).
- Sept. 2022 – Aug. 2023  **EVC-CIN Scholarship 2021.** Grant to Stimulate Scientific Interest in Undergraduate Students.

Honors

- 2023 – 2024  **First Aid to Flag Bearer**, National University of Río Cuarto.

Talks and Presentations

- 2026  **Automatic Inference of Expressive Metamorphic Relations for Effective Metamorphic Testing**
Presented at the Doctoral Symposium of the IEEE International Conference on Software Testing, Verification and Validation (ICST) 2026.
-  **Improving Dynamic Specification Inference with LLM-Generated Counterexamples**
Invited talk, Argentine Workshop on Fundamentals for the Automatic Analysis and Construction of Software (FACAS) 2026, La Falda, Argentina.
- 2022  **Abstraction-aware Inference of Metamorphic Relations**
Invited talk, Argentine Workshop on Fundamentals for the Automatic Analysis and Construction of Software (FACAS) 2022, La Falda, Argentina.

Supervision

- 2025 – Present  **Advisory Collaborator**, Undergraduate thesis,
LLM-based techniques for improving dynamic specification inference.
By Agustín Balestra.
- 2024 – 2025  **Co-advisor**, Undergraduate thesis,
Construction and Integration of Automatic Generators of Complex Inputs in Libraries Supporting Property-Based Testing.
By Augusto Fungo.

Other Academic Activities

- 2025 – Present  **Website & System Administrator**, Department of Computer Science, National University of Río Cuarto, Argentina.